Quick Start Guide
Constant Fraction Discriminator
CFD-X02-A

1 About This Guide

This document is a condensed quick-start guide intended to support basic installation and operation of the device. This guide does **not** replace the full product manual. The complete operation manual can be downloaded at: https://www.flimlabs.com/products/CFD/

Scan the QR code to open the full manual.

Reading the complete manual is strongly recommended before operating the device, unless the user is already familiar with similar laboratory instrumentation.

2 Safety Information

Do not operate the device if it is damaged or if safe operating conditions cannot be ensured.

- Read the full manual before installation and use.
- Operate only within the specified electrical and environmental limits.
- Do not open the enclosure.
- Disconnect power before modifying connections.
- Use only undamaged and properly rated cables.

3 Intended Use

The Constant Fraction Discriminator is intended for use in scientific laboratories and research environments as a timing discriminator for fast analog pulses. The device converts analog detector signals into precisely timed digital pulses suitable for time-correlated measurements and photon counting applications.

The device must be operated only by qualified personnel familiar with laboratory electronic instrumentation.

4 Product Overview

The Constant Fraction Discriminator is a compact electronic module implementing the constant fraction discrimination principle. It converts analog input pulses into stable digital timing signals with minimal timing walk.

The device is optimized for high-speed photon timing applications such as spectroscopy, fluorescence lifetime measurements, and time-correlated photon counting systems.

Key Features

- Constant fraction timing discrimination
- Single analog input channel
- Four simultaneous digital outputs
- Adjustable input threshold
- Adjustable output pulse width
- Monitor output for signal diagnostics
- Compact laboratory form factor

5 Components Overview

Overview of the key components of the device.

1. Polarity selector
2. SMA input
3. Input threshold potentiometer
4. Test point
5. SMA monitor output
6. LED power indicator
7. Output pulse width potentiometer
8. Digital outputs 1 (SMA / USB-C LVDS)
9. Digital outputs 2 (SMA / USB-C LVDS)
10. Power input (USB-C)
11. Delay line connectors (SMA)
12. Grounding point

6 Scope of Delivery

- CFD module
- SMA delay-line cable
- USB Type-C power cable
- Threshold adjustment screwdriver
- Documentation

7 Setup

Install the device on a stable laboratory surface. Before installation:

- inspect the device and accessories for visible damage
- verify connectors and cables are intact
- verify that the input signal meets the requirements given in Table 1

8 Connections

Power connection

- Connect the USB Type-C cable to the device power input.
- Connect the other end to a 5 V DC power source capable of supplying at least 1 A (e.g., computer USB port, USB power adapter, or power bank).
- Verify that the power LED illuminates.

Delay line installation

- Install the external SMA delay cable between the delay output and delay input.
- Select the delay length according to the rise time of the input signal. As a guideline, a 30 cm delay cable corresponds to approximately 1 ns delay.
- Proper delay selection is essential for optimal constant fraction discrimination.

Input signal connection

- Connect the analog input signal to the SMA input port.

Polarity selection

- Set the polarity selector switch according to the polarity of the input signal (positive or negative).

Monitoring connection

- Connect the monitor output and one digital output to an oscilloscope for adjustment and verification.

9 Configuration

After connecting the device:

- verify that the power LED is illuminated
- check that the analog input signal is visible on the monitor output
- confirm that a stable digital output pulse is generated

Adjust the following parameters for optimal device operation:

Input Threshold

The input threshold determines the discrimination level of the analog input signal. Adjustment is performed using the input threshold potentiometer while observing the monitor output and digital output on an oscilloscope. The threshold should be set such that:

- The monitor signal crosses zero cleanly.
- A stable digital output pulse is generated.
- Noise-induced false triggering is avoided.

Output Pulse Width

The output pulse width can be adjusted between approximately 5 ns and 250 ns using the output pulse width potentiometer.

The pulse width must be selected according to the requirements of the connected downstream electronics. Excessively long pulses may reduce the maximum usable repetition rate of the system.

10 Operation

Once the power supply, input signal, and output connections have been correctly established and the initial configuration has been completed, the CFD operates autonomously without further user intervention. An illustration of signal input and output is show in Figure 1.

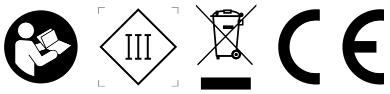
11 Further Information

Detailed information on troubleshooting, maintenance, storage, and disposal is provided in the full operation manual available at https://www.flimlabs.com/products/CFD/.

12 Product Label

Constant Fraction Discriminator

P/N: CFD-X02-A
S/N: CFD-XXXX-XXX
5 V ⎓ 0.5 A | IP40



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13 Technical Specifications

Parameter	Value
Electrical Specifications	
Power supply	5 V DC
Maximum current	0.5 A
Maximum power	2.5 W
Input Signal Requirements	
Input connection	SMA
Input connection	SMA
Input pulse width	> 700 ps
Rise time	> 350 ps
Minimum input amplitude	100 mV
Maximum input amplitude	5 V
Timing jitter	<15 ps
Time walk	±100 ps[a]
Physical Specifications	
Dimensions	87 mm × 87 mm × 40 mm
Weight	225 g
Protection rating	IP40
Environmental Requirements	
Operating temperature	0°C to +40°C
Storage temperature	−20°C to +70°C
Humidity	Up to 80% non-condensing

Table 1: Key specifications of the Constant Fraction Discriminator.

[a] measurement conditions: input pulse amplitude = -100 mV to -4.5 V, fall time = 1 ns, pulse width = 2.5 ns, CFD delay = 1 ns

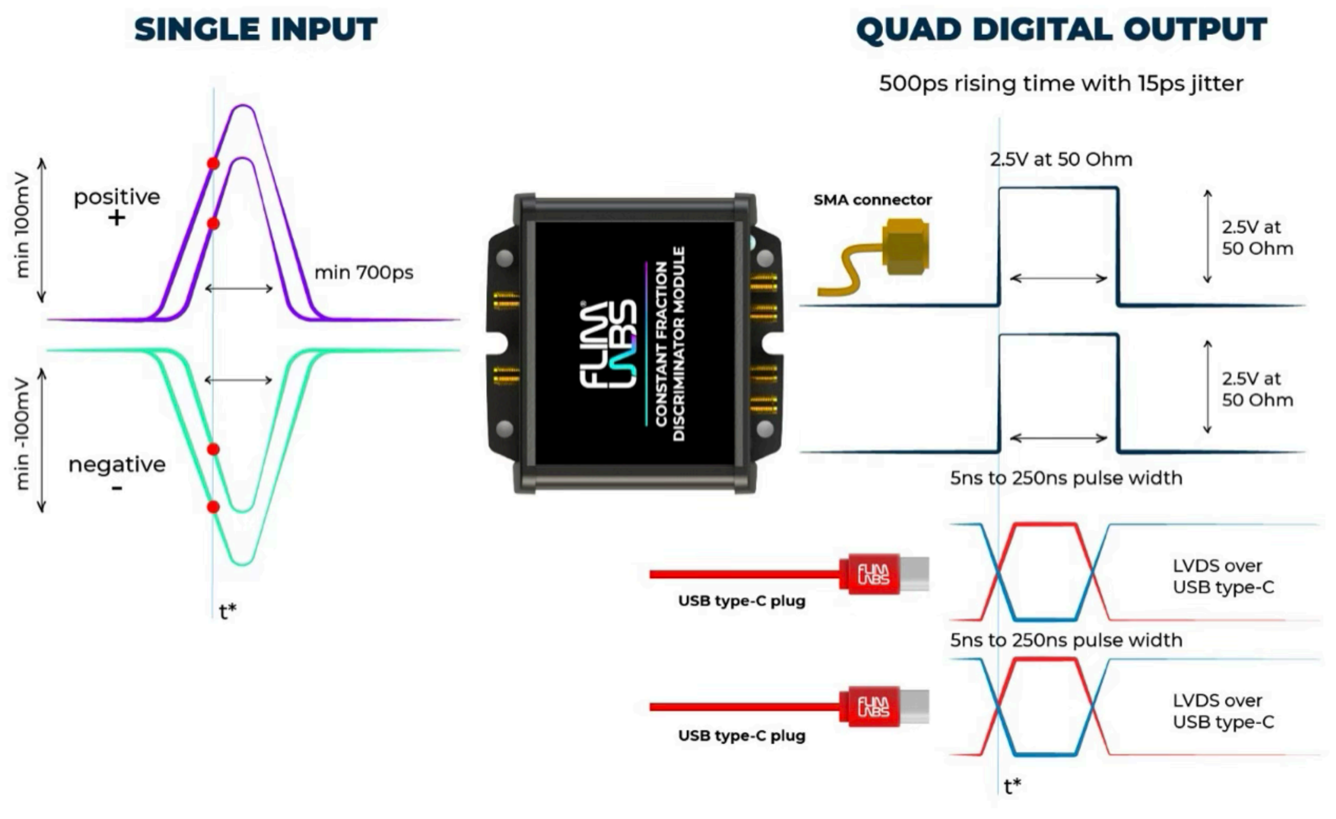

Figure 1: Input and output signal of a configured CFD